

Yuni Wu

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Summary

Data Engineer with a background spanning software engineering, machine learning, and cloud-native data systems. Focused on building trusted data products through validation, monitoring, observability, and metadata design so teams can transform complex operational data into reliable datasets and actionable insights.

Technical Skills

Languages: Python, SQL, Go, R

Data Engineering: ETL/ELT, schema validation, relational models, metadata pipelines, batch/stream ingest, warehouse workflows

Data Quality: anomaly detection, validation checks, regression monitoring, data integrity, root-cause analysis, observability

Cloud & Databases: BigQuery, PostgreSQL, MongoDB, GCP (Cloud Run, Pub/Sub, Cloud Storage), Docker

Systems: FastAPI, REST APIs, asynchronous processing, event-driven workflows, query optimization, automated testing

Experience

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Engineered ROS/rosbag data pipelines to curate 97K+ Unitree Go1 state-action samples for reproducible evaluation datasets.
- Benchmarked behavior-cloning policies across autonomous navigation and human-interaction scenarios.
- Evaluated VLM-guided navigation strategies for decision consistency, semantic understanding, and deployment robustness.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Implemented monitoring, regression validation, and structured logging workflows for Cloud Run inference evaluation.
- Built evaluation pipelines with cross-validation, ROC-AUC, calibration, and validation checks to measure model reliability.
- Evaluated deployment tradeoffs through ablation workflows, reducing inference latency by 5x from 0.15s to 0.03s per image.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered SQL validation and anomaly-detection pipelines for 800+ production scenarios, reducing manual QA by 80%.
- Built regression monitoring pipelines to surface data inconsistencies faster, accelerating reporting speed by 78%.
- Optimized distributed ETL workflows and GCP data-access patterns, reducing query latency by 15%.
- Automated API/Selenium data collection for validation, monitoring, and reliability analysis across production systems.

Projects

NailAI — Cloud-Native Metadata and Data Pipeline Platform

[demo]

- Built event-driven processing workflows using Cloud Run, Pub/Sub, and Cloud Storage for asynchronous image inference.
- Designed request, job, and prediction metadata pipelines for BigQuery monitoring, diagnostics, and analytics.
- Achieved 0% failure at 200 concurrent users with p95 latency of 6.8s using idempotent orchestration and retry handling.
- Optimized BigQuery partitioning and clustering patterns for warehouse-centric analysis and query efficiency.

FitMatch — Product Data Modeling and Validation Platform

- Built a FastAPI/Next.js data product with PostgreSQL schemas, catalog ingestion, product mappings, and QA tooling.
- Validated catalog data across 1,205 products, 65 size profiles, 407 chart rows, and 1,205 product-profile relationships.
- Implemented validation workflows, feedback collection, and regression-style boundary tests, achieving 24/24 passing cases.

Stock Intelligence Copilot — API-Backed Data Product

[demo]

- Built FastAPI/Next.js workflows combining market data, fundamentals, news, earnings, and portfolio signals.
- Implemented evidence aggregation and traceable output contracts with per-tool error capture and fallback-safe API responses.
- Expanded automated testing across runtime execution, tool contracts, route adapters, portfolio scenarios, and streaming behavior.

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0

May 2026

Soochow University — BS, Data Science | GPA: 3.7/4.0

June 2021